

CS 421: SOFTWARE DEPLOYMENT (2025/2026)

REAL-TIME COMMUNICATION INTEGRATION (BODABODA APP)

(Extension of CI/CD Assignment)

1. Objective of Tasks

In your previous assignment, you:

- Containerized your Bodaboda application using Docker
- Built a CI/CD pipeline for automated testing and deployment

In assignment, you will:

- Introduce **real-time communication using MQTT**
- Transform your system into an **event-driven application**
- Integrate MQTT into your **existing Docker and CI/CD setup**

2. What is expected: To enhance your system by adding:

A **messaging layer (MQTT)** that allows components to communicate in real time.

3. REQUIRED IMPLEMENTATION

You are required to implement ANY ONE (1) of the following features using MQTT: (You may also implement all the features)

Option A: Ride Request Broadcasting

- Passenger sends ride request
- Backend publishes message to MQTT topic
- Driver (subscriber) receives request instantly

Option B: Driver Location Updates

- Driver continuously publishes location
- Passenger subscribes and receives updates in real-time

Option C: Ride Status Updates

- Driver publishes:
 - ride accepted
 - ride started
 - ride completed
- Passenger receives updates instantly

4. TASK INSTRUCTIONS

TASK 1: SET UP MQTT BROKER

- Add an MQTT broker to your system using Docker (Use any of your favorite broker)

Example:

```
docker run -d -p 1883:1883 eclipse-mosquitto
```

TASK 2: DEFINE YOUR MESSAGE DESIGN

You must clearly define:

- At least **one topic**, e.g.:

ride/request

driver/location

ride/status

- Message format (JSON preferred), e.g.:

```
{
  "user_id": 1,
  "location": "Nyerere Square"
}
```

TASK 3: MODIFY YOUR BACKEND

Update your application to:

- **Publish messages** to MQTT broker
- **OR Subscribe to messages** depending on your feature

TASK 4: CREATE A SIMPLE CLIENT SIMULATION

You must demonstrate:

- A **publisher** (e.g., passenger or driver)
- A **subscriber** (e.g., driver or passenger)

This can be:

- A simple script
- Another service
- Or part of your app

TASK 5: UPDATE DOCKER SETUP

Extend your existing setup to include:

- Application container
- MQTT broker container

Preferably use:

docker-compose.yml

TASK 6: INTEGRATE INTO CI/CD PIPELINE

Update your pipeline to:

- Start MQTT broker during deployment
- Ensure application connects successfully
- (Optional Advanced) Add a simple test:
 - Publish → Receive message

5. DEMONSTRATION REQUIREMENTS

During presentation, your group must show:

- MQTT broker running
- Messages being sent and received
- Real-time system behavior

6. SUBMISSION REQUIREMENTS

A. CODE SUBMISSION

- Updated GitHub repository including:
 - MQTT integration
 - Docker configuration
 - CI/CD updates

B. DIGITAL REPORT (ADD SECTION)

Add a new section:

“MQTT Integration”

Include:

- Feature implemented

- Topics used
- Message format
- Screenshots of message exchange
- Explanation of how it works

C. HANDWRITTEN DOCUMENT (INDIVIDUAL)

Each student must explain:

- What is MQTT
- Difference between HTTP and MQTT
- How MQTT improved your Bodaboda system
- One real-world application (e.g., Uber, Bolt)

DEADLINE: 28/5/2026, MUDA WA KIPINDI